Job Market Analysis using SQL and Power BI

Complete Project Submission Package

📁 FOLDER STRUCTURE

```
Job_Market_Analysis/
│
├── 📁 Dataset/
│   └── jobs_dataset.csv
│
├── 📁 SQL/
│   └── queries.sql
│
├── 📁 PowerBI/
│   └── dashboard_Job_analysis.pbix
│
├── 📁 Presentation/
│   └── Job_Market_Analysis_Presentation.pptx
│
├── 📁 Report/
│   └── Project_Report.pdf
│
├── 📁 Screenshots/
│   ├── dashboard_page1.png
│   └── dashboard_page2.png
│
└── README.md
```

📄 README FILE

Job Market Analysis — Data Analyst Internship Project

Project: Job Market Analysis using SQL and Power BI
Type: Data Analyst Internship Project
Year: 2026

Overview

This project analyzes job market trends across the United States for data science-related roles. Using a dataset of 742 job postings (42 features), I extracted and cleaned the data using SQL, then built an interactive Power BI dashboard to uncover hiring patterns, salary trends, skill demand, and the impact of education on compensation.

Tools Used

- **SQL** — Data extraction and initial cleaning
- **Power BI** — Dashboard development and visualization
- **Power Query** — Additional data transformations within Power BI
- **DAX** — Calculated measures for KPI metrics

Files Included

File	Description
`jobs_dataset.csv`	Raw/cleaned dataset used for analysis
`queries.sql`	All SQL queries used for extraction and aggregation
`dashboard_Job_analysis.pbix`	Interactive Power BI dashboard file
`Job_Market_Analysis_Presentation.pptx`	Final presentation slides
`Project_Report.pdf`	Written project report
`Screenshots/`	Dashboard screenshots for reference

How to Open

1. Open `dashboard_Job_analysis.pbix` in Power BI Desktop
2. Use the slicers (Job Title, State) to filter the dashboard
3. Navigate between pages using the tabs at the bottom
4. Refer to `queries.sql` for the SQL logic used prior to Power BI import

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📝 PROJECT REPORT

Introduction

This project was completed as part of a Data Analyst Internship. The goal was to analyze the data science job market across the United States using real-world job posting data. The dataset contained 742 rows and 42 columns, covering job titles, salary estimates, company details, industries, required skills, education requirements, and location information.

The analysis was carried out using SQL for data extraction and initial cleaning, followed by Power BI for transformation and interactive dashboard development.

Data Cleaning

The raw dataset required a few rounds of cleaning before it was ready for visualization.

In SQL:

- Identified and removed duplicate records
- Standardized state abbreviations and company name formatting

- Filtered out records with missing or incomplete job titles

In Power Query (Power BI):

- Replaced `-1` values in the salary columns with `null` — these were placeholder values in the original dataset indicating "not disclosed"
- Split the salary range field into separate `lower_salary` and `upper_salary` columns
- Renamed columns for clarity (e.g., `avg_salaryK` → readable label)
- Removed rows where both salary fields were null after the replacement

After cleaning, the dataset was ready for analysis without significant data loss.

Dashboard Overview

The Power BI dashboard is split into two pages:

Page 1 — Salary & Job Distribution

- 4 KPI cards: Total Jobs, Total Companies, Total Industries, Avg Salary
- Bar chart: Avg Max and Min Salary by State
- Bar chart: Salary by Job Title
- Pie chart: Top 5 Industries by Job Count
- Bar chart: Job Count by State
- Treemap: Top 10 Companies by Job Openings
- Slicers: Job Title, State

Page 2 — Skills & Education

- KPI cards: Avg Rating, Most Demanded Skill, Highest Paying Role, Total States
- Pie chart: Average Salary by Degree Level
- Map visual: Average Salary by State
- Matrix table: Skills Required by Companies for Each Job Title

Both pages are interactive — slicers on either page update all visuals simultaneously.

Key Insights

1. **California dominates** in both volume (98 job postings) and salary ($157K avg max).
2. **Data Scientist** is the most listed role, representing the largest share of postings.
3. **Python and SQL** are required in the overwhelming majority of listings — non-negotiable skills.
4. **Biotech & Pharmaceuticals** is the top hiring industry with 24 postings, ahead of Enterprise Software (19).
5. **PhD holders** earn an average of $134K, compared to $101.5K for roles with no degree requirement — a 32% gap.
6. **Machine Learning Engineers** consistently appear in the top salary bracket across states.
7. **MassMutual** is the top company by job openings (8 postings), followed by Takeda Pharmaceuticals (7).
8. **AWS, Spark, and Tableau** form the secondary skills tier after Python and SQL.
9. **Massachusetts and New York** are strong hiring hubs outside of California.

10. Company **ratings average 3.72** across all postings — signaling a competitive but not exceptional employer landscape.

Recommendations

For Job Seekers:

- Focus on Python and SQL first — they appear in nearly every listing.
- Target California, Massachusetts, and New York for the highest volume of opportunities.
- Pursuing a Masters or PhD can meaningfully increase expected salary.
- Add cloud skills (AWS) and big data tools (Spark) as secondary differentiators.

For Hiring Companies:

- Expand pipelines beyond California to reduce competition and find strong talent in Virginia, Illinois, and Maryland.
- Salary transparency in job descriptions improves applicant quality.
- Partnering with universities in high-concentration states (Massachusetts) can build strong talent pipelines.

Conclusion

This project gave me hands-on experience with the full data analysis workflow — from raw database extraction to a polished, interactive dashboard. Working with SQL helped me understand how to structure queries for real analytical questions. Power BI and DAX taught me how to translate those queries into visual KPIs that non-technical stakeholders can actually use.

The job market findings were clear and consistent: Python, SQL, and location (especially California) are the biggest drivers of data science hiring and compensation in the US.

🗄️ SQL QUERIES DOCUMENT

```sql
-- ============================================================
-- Job Market Analysis – SQL Queries
-- Database: jobs
-- Purpose: Data extraction and aggregation for Power BI import
-- ============================================================


-- ------------------------------------------------------------
-- Query 1: Job Distribution by State
-- Shows which states have the most data science job openings
-- Used for: State-level bar chart in dashboard
-- ------------------------------------------------------------
SELECT
    state,
```

```sql
    COUNT(*) AS total_jobs
FROM jobs
GROUP BY state
ORDER BY total_jobs DESC;


-- ------------------------------------------------------------
-- Query 2: Top 5 Industries by Hiring Volume
-- Identifies industries actively recruiting data science roles
-- Used for: Industry pie chart in dashboard
-- ------------------------------------------------------------
SELECT
    industry,
    COUNT(*) AS total_jobs
FROM jobs
GROUP BY industry
ORDER BY total_jobs DESC
LIMIT 5;


-- ------------------------------------------------------------
-- Query 3: Average Salary by Job Title
-- Compares compensation across different data science roles
-- Used for: Salary by job title bar chart
-- ------------------------------------------------------------
SELECT
    job_title,
    AVG(avg_salaryK) AS avg_salary
FROM jobs
GROUP BY job_title;


-- ------------------------------------------------------------
-- Query 4: Average Salary by State (Min and Max)
-- Used for salary range comparison by state
-- ------------------------------------------------------------
SELECT
    state,
    AVG(lower_salary) AS avg_min_salary,
    AVG(upper_salary) AS avg_max_salary
FROM jobs
GROUP BY state
ORDER BY avg_max_salary DESC;


-- ------------------------------------------------------------
-- Query 5: Top Companies by Job Count
-- Identifies which companies are hiring the most
-- Used for: Treemap visual in dashboard
-- ------------------------------------------------------------
SELECT
    company_name,
    COUNT(*) AS total_openings
FROM jobs
```

```sql
GROUP BY company_name
ORDER BY total_openings DESC
LIMIT 10;


-- ------------------------------------------------------------
-- Query 6: Education Level vs Average Salary
-- Analyzes how degree requirements impact compensation
-- Used for: Education pie chart in Page 2
-- ------------------------------------------------------------
SELECT
    min_education AS degree_required,
    AVG(avg_salaryK) AS avg_salary,
    COUNT(*) AS job_count
FROM jobs
GROUP BY min_education
ORDER BY avg_salary DESC;
```

✅ SUBMISSION CHECKLIST

Before submitting, confirm all files are present:

- [ ] `dashboard_Job_analysis.pbix` — Power BI dashboard file
- [ ] `Job_Market_Analysis_Presentation.pptx` — 16-slide presentation
- [ ] `Project_Report.pdf` — Written report
- [ ] `queries.sql` — All SQL queries with comments
- [ ] `jobs_dataset.csv` — Dataset file
- [ ] `Screenshots/dashboard_page1.png` — Dashboard page 1
- [ ] `Screenshots/dashboard_page2.png` — Dashboard page 2
- [ ] `README.md` — Project overview and file guide

💼 PROFESSIONAL PROJECT DESCRIPTIONS

Resume Description

> **Job Market Analysis — Data Analyst Internship Project**
> Analyzed 742 data science job postings using SQL and Power BI to identify hiring trends, salary patterns, and skill demand across 37 US states. Extracted and cleaned data via SQL queries, applied Power Query transformations, and developed DAX measures for KPI calculations. Built a two-page interactive dashboard featuring salary comparisons, industry breakdowns, and an education-vs-salary analysis. Key finding: California leads in both volume and salary, with Python and SQL required in over 95% of postings.

LinkedIn Project Description

> **Job Market Analysis using SQL and Power BI**
>
> Completed as part of my Data Analyst Internship, this project analyzes real-world job market data for data science roles across the United States.
>
> **What I did:** Extracted 742 job postings from a SQL database, cleaned and transformed the data using Power Query, built DAX measures for KPI tracking, and developed a fully interactive Power BI dashboard.
>
> **What I found:** California leads with 98 postings and the highest salaries. Python and SQL are near-universal requirements. PhD holders earn ~32% more than those without a specified degree. Biotech & Pharmaceuticals is the top hiring industry.
>
> **Tools:** SQL · Power BI · Power Query · DAX

GitHub README Description

> # Job Market Analysis — SQL & Power BI
>
> A data analytics project analyzing 742 US data science job postings to uncover hiring trends, salary patterns, and skill demand.
>
> **Stack:** SQL · Power BI · Power Query · DAX
> **Dataset:** 742 rows × 42 columns — job titles, salaries, companies, industries, skills, education, states
>
> **Key Findings:**
>
> - California: highest job count (98) and salary ceiling ($157K avg max)
> - Most demanded role: Data Scientist
> - Top skills: Python (258 mentions), SQL (253 mentions)
> - PhD premium: +32% salary over no-degree roles
> - Top industry: Biotech & Pharmaceuticals (24 postings)
>
> **Project includes:** SQL queries, .pbix dashboard file, presentation, and full report.

🎤 INTERVIEW PREPARATION

Common Interview Questions & Suggested Answers

Q: Walk me through your project.

> "I analyzed a dataset of 742 data science job postings using SQL and Power BI. I started by connecting to the database, running queries to extract and clean the data, then imported it into Power BI. I used Power Query for additional transformations — like replacing -1 salary placeholders — and built DAX measures for KPIs like average salary and total job count. The final dashboard has two pages: one for geographic and salary analysis, and another for skills and education insights."

Q: Why did you use SQL instead of doing everything in Power BI?

> "SQL is better for initial data extraction and heavy aggregation — it runs directly on the database and is faster for things like GROUP BY queries on large tables. Power BI is designed for visualization and reporting, not raw data wrangling. Splitting the work made sense: SQL handles the extraction and cleanup, Power BI handles the presentation."

Q: What is a DAX measure? Can you give an example?

> "A DAX measure is a calculated formula in Power BI that computes dynamically based on filters applied in the report. For example, `Total Jobs = COUNTROWS(Jobs)` counts the number of rows in the dataset — but when a user filters by state or job title using a slicer, the measure automatically recalculates for that selection. It's different from a calculated column because it doesn't get stored in the table."

Q: How did you handle the -1 values in salary columns?

> "The -1 values were placeholder entries in the original dataset indicating the salary wasn't disclosed. I replaced them with `null` using Power Query's Replace Values function so they wouldn't skew the salary averages. Any row where both min and max salary were null after that was excluded from salary-related visuals."

Q: What was the most interesting finding from your analysis?

> "The education-salary gap was pretty striking. PhD holders earn about $134K on average versus $101.5K for roles with no degree requirement — that's a 32% difference. It shows that in data science, advanced education still has a real market premium, especially at research-heavy companies in Biotech and Pharma."

Q: What would you improve if you had more time?

> "I'd add a time-series dimension to track how demand for specific skills or roles is changing over time. I'd also build a predictive salary model using regression, and potentially create a role-matching tool where a user inputs their skills and it suggests which states and industries to target."

Q: What did you learn from this project?

> "The most practical thing was getting comfortable with the full pipeline — from database query to a polished dashboard. I also got much better at DAX, which was new to me. And honestly, the data storytelling part was really valuable: learning to pick the right chart type and present findings in a way that makes sense to someone who hasn't seen the data before."

Project completed as part of a Data Analyst Internship · 2026